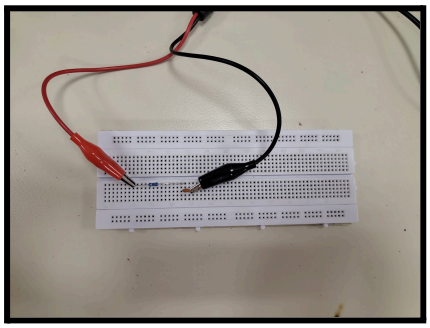
Sound Filtration Systems

Analysis of Low-Pass, High-Pass, and Tone Control Audio Filters

1. Low-Pass Filter and Frequency Response

1.1 Low-Pass Filter Circuit Architecture

The simplest form of a low-pass filter is the RC circuit shown below.
For R = 15kΩ and C = 2.2nF, the following circuit is constructed.



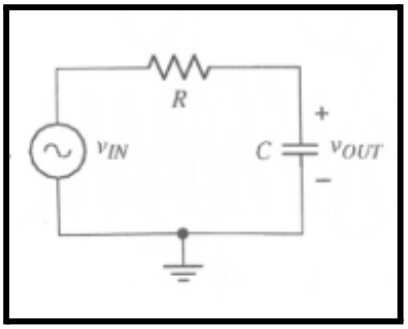
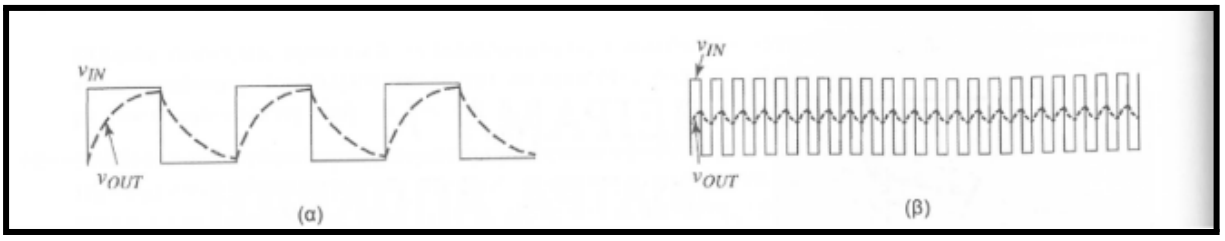


Figure 1 - Filter circuit schematic and implementation

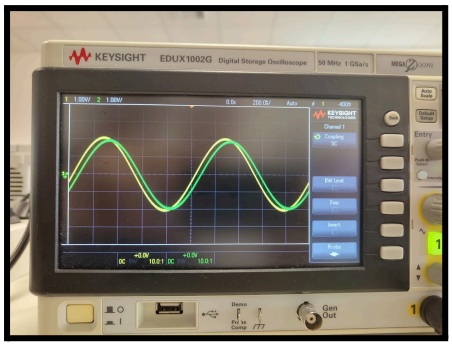
For low signal input frequencies, the capacitor has sufficient time to charge and discharge, and therefore the output closely mimics the input. As the frequency increases, the capacitor can neither fully charge nor discharge, resulting in waveforms of extremely low amplitude that approach a constant zero value. The two cases are observed in the following figure.



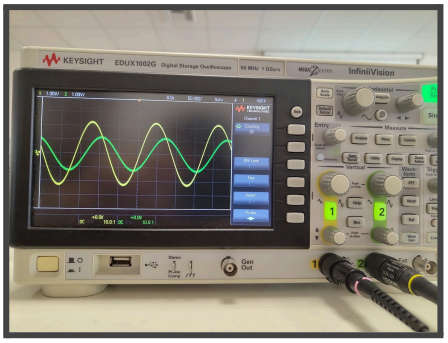
(a) low frequencies (b) high frequencies

1.2 Input/Output Waveforms across Frequencies

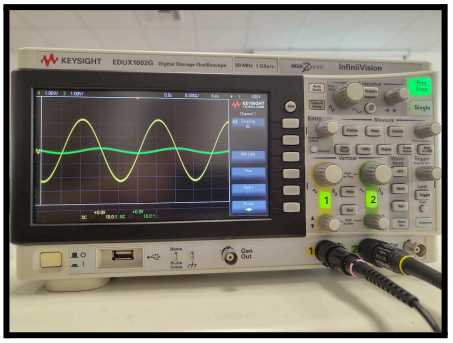
The figure below illustrates the input (yellow) and output (green) waveforms for frequency values f = 1, 6, and 50 kHz for a sinusoidal Vin with an amplitude of 2V.



f = 1kHz



f = 6kHz



f = 50kHz

1.3 Cutoff Frequency: Theoretical and Experimental Analysis

The cutoff frequency is defined as the critical frequency value that separates the voltage attenuation region from the passband region where the output follows the input.

Theoretically, it is the frequency at which the angular frequency ω equals the reciprocal of the time constant τ. For the above circuit, the following relations hold:

- $\omega = 2\pi f$
- $\tau = RC$

Therefore, for the given component values: $f_c = \frac{1}{2\pi RC} \Rightarrow fc = 4823\ Hz$

Experimentally, the measurement procedure is as follows:

- It is known that for $f = f_c$, the gain $G = \frac{A_{out}}{A_{in}}$ equals $\frac{1}{\sqrt{2}}$.
- Therefore, for $f = f_c$, $A_{in} = 2V \Rightarrow A_{out} = \sqrt{2}\,V$

By varying the frequency and observing the output on the oscilloscope, the cutoff frequency is measured as $f_c = 4960\ Hz$, accurate to two digits of $\sqrt{2}$.

The observed error stems from the finite representation of irrational numbers as well as systematic instrument errors.

To adjust the cutoff frequency, assuming C = 2.2nF and using the theoretical frequency value, we get: $f_c = \frac{1}{2\pi RC} \Rightarrow R = \frac{1}{2\pi C f_c} = 15\Omega$

1.4 Amplitude Measurements and Bode Magnitude Plot

Laboratory measurements yield the following table of data:

f (kHz)	Ain (V)	Aout (V)	G - Aout/Ain	G (dB)
0.01	2	1.97	0.985	-0.13
0.1	2	1.97	0.985	-0.13
0.3	2	1.97	0.985	-0.13
1	2	1.83	0.915	-0.77
3	2	1.75	0.875	-1.16
10	2	0.91	0.455	-6.84
100	2	0.17	0.085	-21.41

Table 1 - Amplitude, Frequency, and Gain Measurements

Table 1 presents the gain G in dB scale, calculated according to the formula: $G = 20log_{10}(\frac{A_{out}}{A_{in}})$

From the table measurements, the Bode magnitude plot is derived:

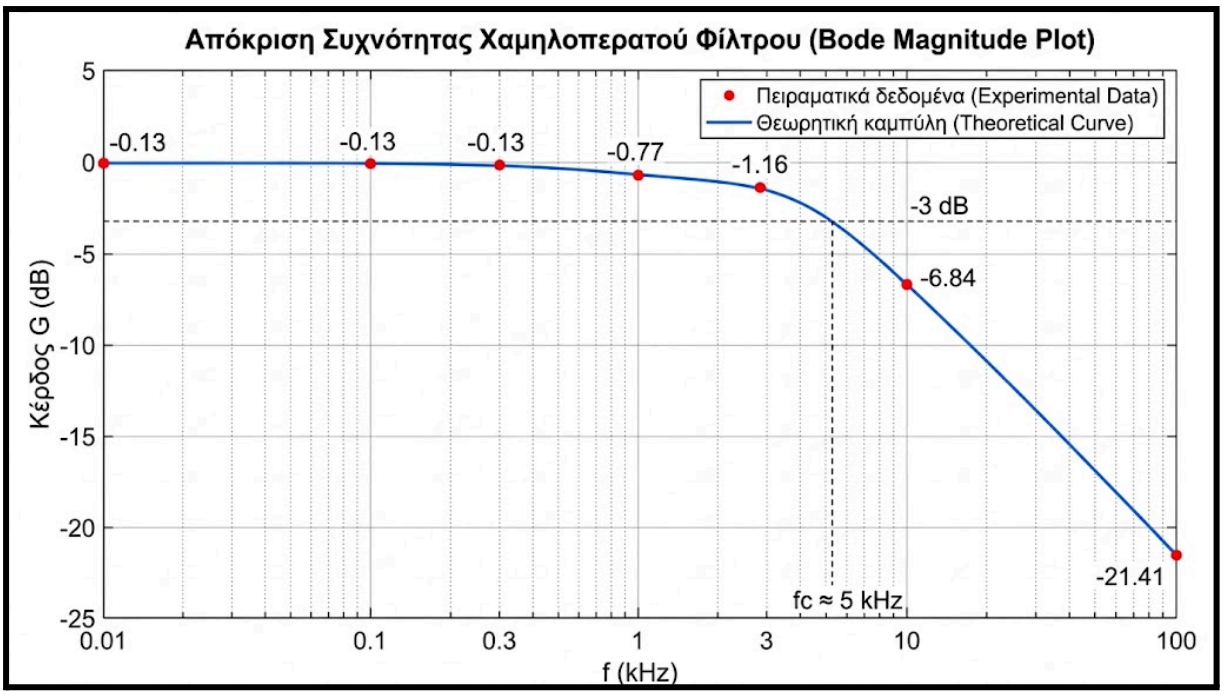


Figure 3 - Bode magnitude plot on logarithmic frequency scale

1.5 Phase Shift Characteristics and Bode Phase Plot

As observed in Figure 1, there is a noticeable phase difference between the input and output signals of the circuit. Theoretically, this phase shift is given by the formula $\varphi = -\frac{\Delta t}{T}360^o$, where Δt represents the time delay between the two signals.

Specifically, at the cutoff frequency, the phase shift is φ = -45°.
Experimentally, it is calculated by measuring Δt, yielding:

$$\varphi = -\frac{\Delta t}{T}360^o \Rightarrow \varphi = -f\Delta t 360^o = -48.64$$

Taking additional representative frequency values produces the following table:

f (kHz)	0.01	0.1	0.3	1	3	10	100
φ	0.68	1.97	6.04	16.34	37.33	67.4	77.48

Table 2 - Frequency vs. Phase Shift

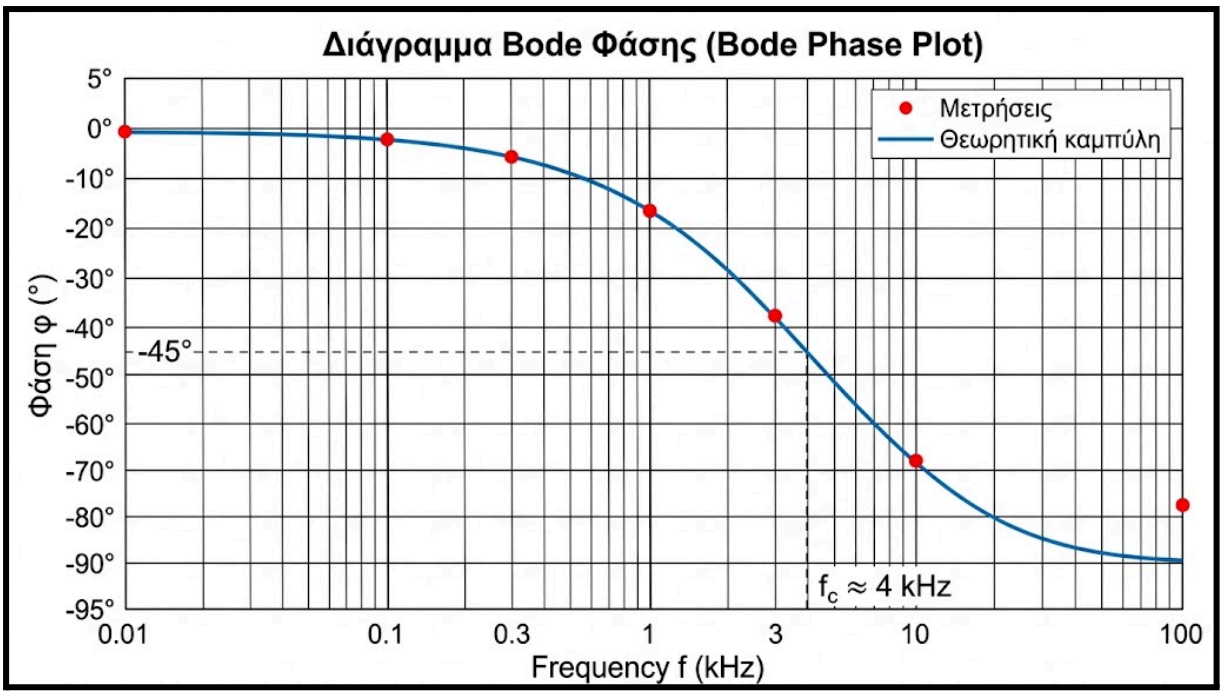


Figure 4 - Bode phase plot

2. High-Pass Filter and Frequency Response

2.1 High-Pass Filter Circuit Architecture

The high-pass filter circuit is constructed by considering Vout as the voltage across the resistor. For this implementation, R = 15kΩ and C = 2.2nF are used.

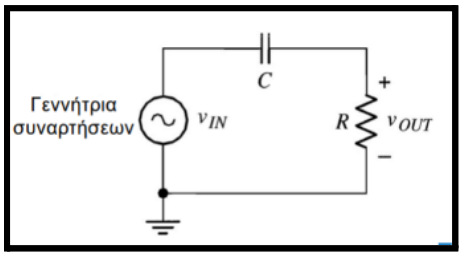


Figure 5 - High-pass filter schematic

In the high-pass filter configuration, the operation is inverted. At high frequencies, the capacitor acts as a short circuit, and the parallel connection between generator and resistor ensures Vin = Vout. Conversely, at low frequencies where the capacitor acts as an open circuit, current is blocked, resulting in Vout = 0.

For representative frequency values, the input-output waveforms are as follows:

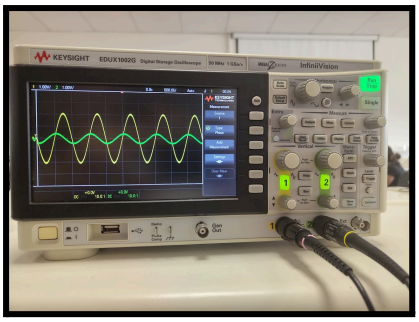
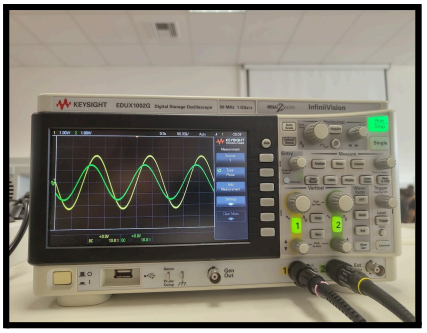
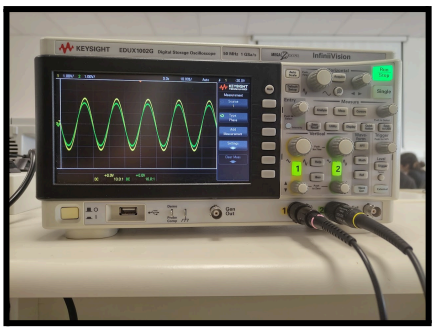




Figure 6 - Oscilloscope waveforms

f = 1kHz f = 6kHz f = 50kHz

2.2 Cutoff Frequency Calculation and Measurement

Correspondingly to the low-pass filter, the cutoff frequency is defined for the high-pass filter as well. The equation and theoretical value remain identical, while the experimental value is determined to be 4575 Hz using an equivalent methodology. Deviation occurs for the same reasons as in the low-pass case. Tuning to the exact cutoff frequency requires the same resistor value adjustment to 15kΩ.

2.3 Amplitude Response and Bode Magnitude Plot

f (kHz)	Ain (V)	Aout (V)	G - Aout/Ain	G (dB)
0.01	2	0.1	0.05	-26.02
0.1	2	0.12	0.06	-24.44
0.3	2	0.2	0.10	-20.00
1	2	0.47	0.235	-12.58
3	2	1.1	0.55	-5.19
10	2	1.82	0.91	-0.82
100	2	2	1.00	0.00

Table 3 - High-Pass Filter Gain vs. Frequency

The Bode magnitude plot derived from the table data is shown below:

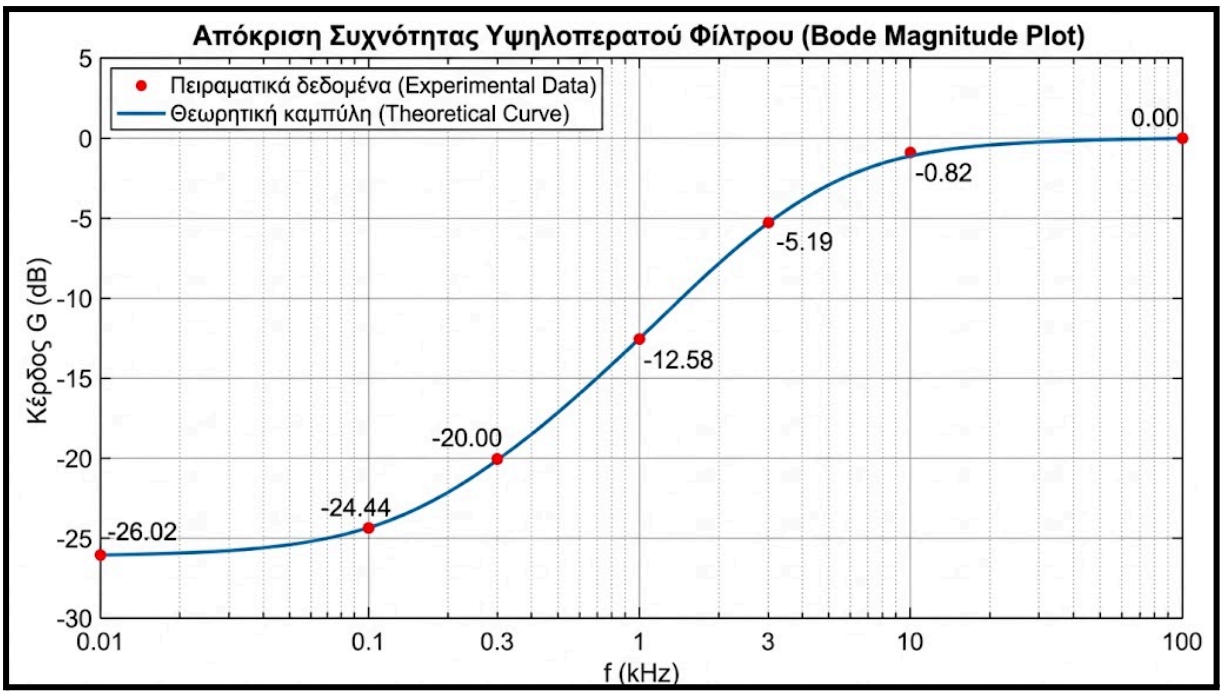


Figure 7 - Bode magnitude plot for high-pass filter

3. Audio Tone Control System Integration

3.1 Audio Amplification System: Unfiltered vs. Filtered Configurations

3.1.1 Circuit Configuration Without Filter

The following configuration forms an audio amplifier. It receives a weak input signal, conditions it, removes any DC bias, and amplifies it (by a gain factor of 11), driving the speaker to output clear, amplified audio with a near-square waveform.

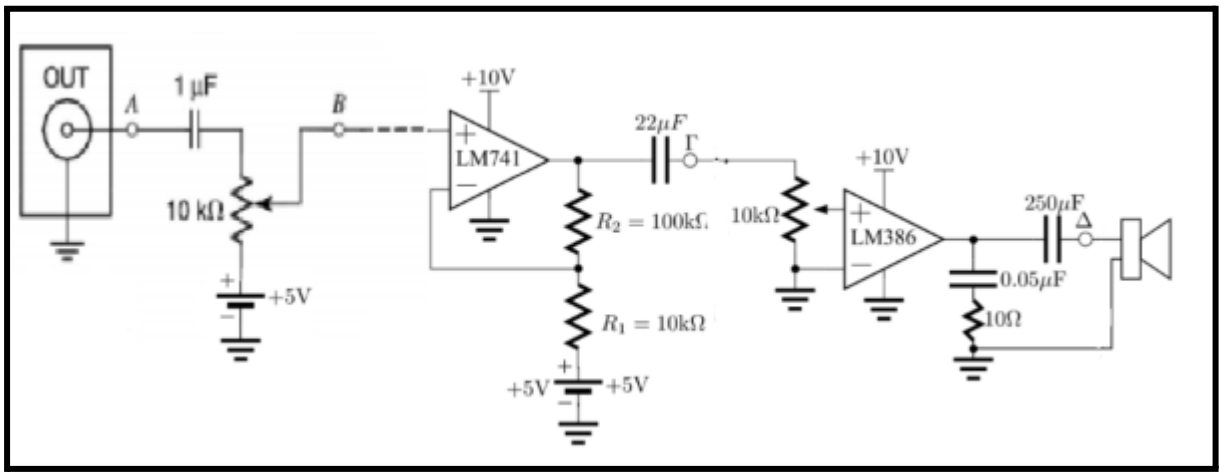
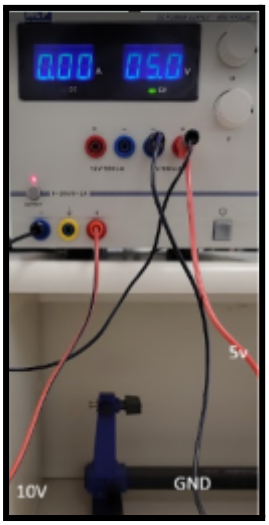


Figure 8 - Unfiltered system schematic diagram



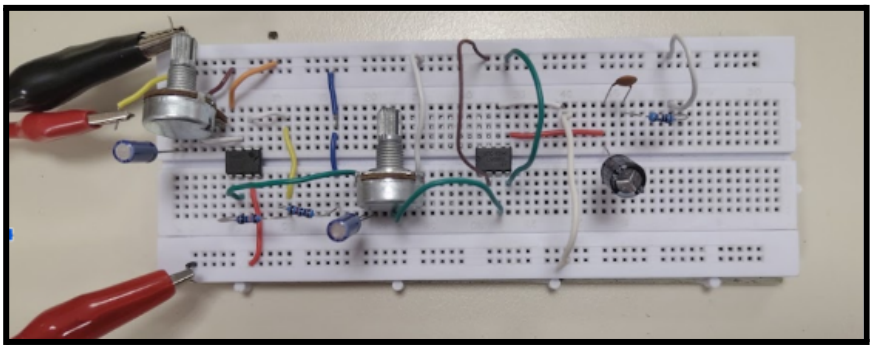


Figure 9 - Hardware implementation (Additional wiring was added across the breadboard power rails to distribute GND and supply voltages across the full board length)

3.1.2 Circuit Configuration With Filter

Incorporating the low-pass filter does not alter the fundamental operation of the circuit; however, it restricts its operational bandwidth. The low-pass filter establishes a cutoff frequency: for any frequency below this threshold, the circuit cleanses and amplifies the signal, whereas frequencies above the cutoff frequency are attenuated/suppressed.

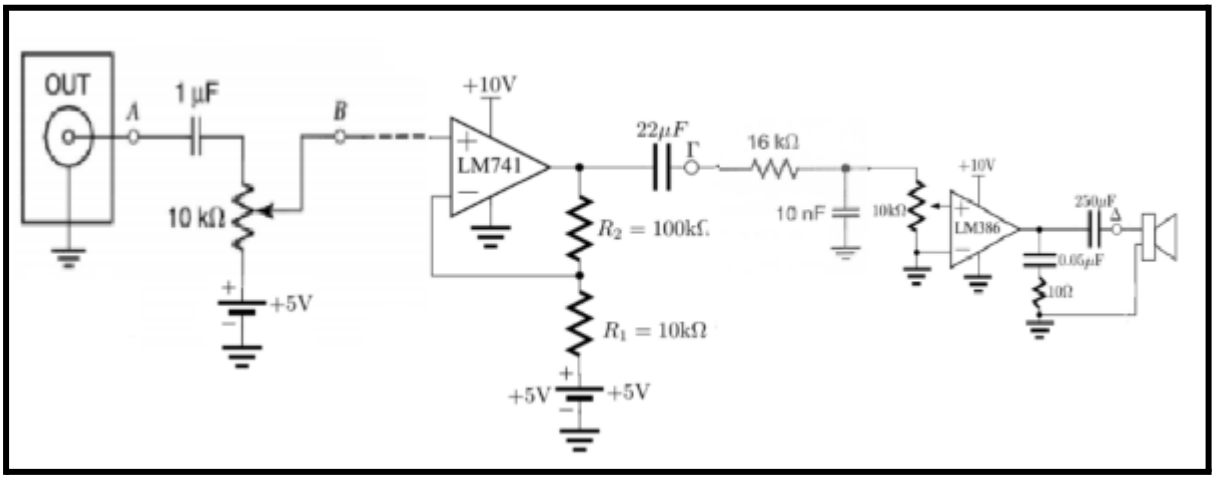
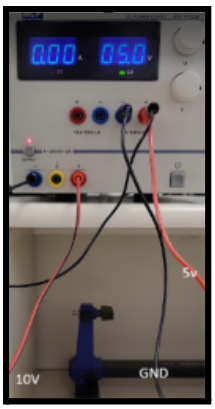


Figure 10 - Filtered system schematic diagram



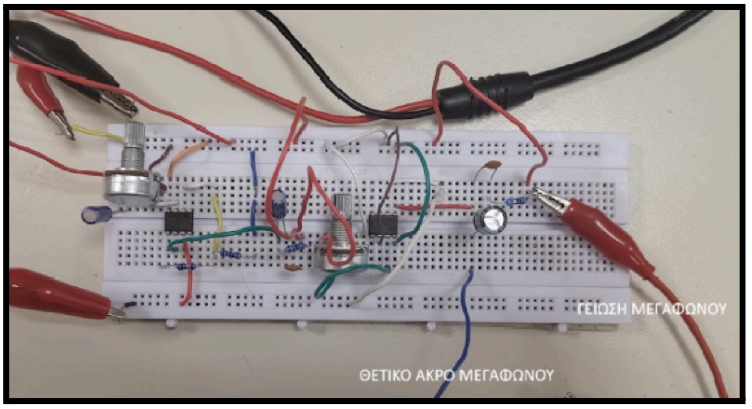


Figure 11 - Filtered system implementation

3.1.3 Experimental Comparison and Observations

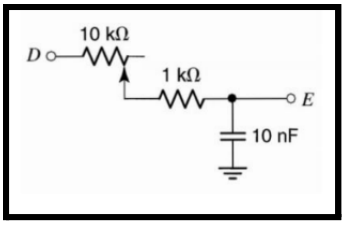
Testing representative input frequencies across the 20 Hz – 20 kHz audio spectrum highlights the following differences between the two configurations:

Circuit A (Unfiltered)	Circuit B (Filtered)	Technical Explanation
Constant signal amplification across the entire frequency range	Signal amplification restricted to f < 1 kHz	1 kHz corresponds to the cutoff frequency of Circuit B
Stable amplification magnitude	As frequency approaches the cutoff point, amplification progressively weakens	Capacitor lacks adequate charging/discharging time per cycle
Input and output signals remain in-phase	Progressive phase shift between input and output	Low-pass filtering inherently introduces frequency-dependent phase lag
Clear, powerful acoustic output from the loudspeaker	Muffled, attenuated sound as pitch rises, leading to total attenuation	Filter frequency attenuation directly shapes acoustic audio performance

Table 4 - Comparative Performance Observations

3.2 Variable Resistance Filters (Tone Control Tuning)

For the variable resistance low-pass filter shown below, the following relations apply:

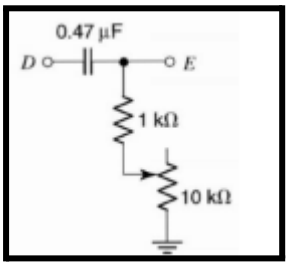


$$1 \leq R_{o\lambda} \leq 11 \Rightarrow \frac{1}{11} \leq \frac{1}{R_{o\lambda}} \leq 1 \Rightarrow$$

$$\frac{1}{22\pi C} \leq f_c \leq \frac{1}{2\pi C} \Rightarrow 1.45kHz \leq f_c \leq 15.92kHz$$

Figure 12 - Variable resistance low-pass filter topology

For the variable resistance high-pass filter shown below, the following relations apply:



$$1 \leq R_{o\lambda} \leq 11 \Rightarrow \frac{1}{11} \leq \frac{1}{R_{o\lambda}} \leq 1 \Rightarrow$$

$$\frac{1}{22\pi C} \leq f_c \leq \frac{1}{2\pi C} \Rightarrow 30.78kHz \leq f_c \leq 338.63kHz$$

Figure 13 - Variable resistance high-pass filter topology

The high-pass filter attenuates the input signal for frequencies f < fc. Consequently, increasing Rtotal decreases fc, thereby broadening the passband frequency range enabled for audio amplification.

Note: The present document is an AI-translated version of a lab report as part of the Introductory Laboratory of Electronics and Telecommunication course at NTUA.
The lab followed the steps of Exercise 7 from the book ‘Introductory Laboratory of Circuits and Electronics’ , by Yannis Tsividis.